The empirical recurrence for OEIS A233679, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A233679 records “Empirical recurrence of order 99” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 4096$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the whole sequence, so the recurrence holds from $n = 100$ — every index at which it can be written down. Its last coefficient is $c_{99} = -5316911983139663491615228241121378304 \neq 0$, so no further tail satisfies anything shorter; any order-99 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A233679 is “Number of (n+1)X(5+1) 0..3 arrays with every 2X2 subblock having the sum of the squares of the edge differences equal to 6.”. It has offset 1 and begins

$$8712, 117744, 1500800, 25036784, 365293440, 6336415296, 95799636288, 1660462199232, \ldots$$

The entry states:

> Empirical recurrence of order 99 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 08:57:08, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 4096$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 4096$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Here the stated order is already the minimal order of the sequence itself, so no tail has to be taken.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 8192$ exact terms from $n = 1$ on, it returns order 99 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{99} c_i \, a(n-i),$$

c_1	8	c_{26}	36483962261145968720543744	c_{51}	−80972430518518
c_2	891	c_{27}	−296611304439334781723181056	c_{52}	−23028456357031
c_3	−6878	c_{28}	−792895795296705501963157504	c_{53}	394253842906453
c_4	−355072	c_{29}	6618144808412560656766599168	c_{54}	880683008473595
c_5	2680092	c_{30}	14918571597917552242997067776	c_{55}	−170096640183935
c_6	85525456	c_{31}	−128288484171147870988605587456	c_{56}	−29258609698988
c_7	−636643736	c_{32}	−243980651056340128888506023936	c_{57}	6498188925403685
c_8	−14119175912	c_{33}	2169567370888981396593903140864	c_{58}	838871738344238
c_9	104294068320	c_{34}	3479435768518746080327686422528	c_{59}	−219586433241047
c_{10}	1712695457888	c_{35}	−32124552415195720460375785734144	c_{60}	−205471297295669
c_{11}	−12614786644864	c_{36}	−43383255626868345796311283400704	c_{61}	6554029421073354
c_{12}	−159555598168960	c_{37}	417713292660431811003430828244992	c_{62}	4228570749725709
c_{13}	1176562161629952	c_{38}	473895977562080073021265455087616	c_{63}	−1724620669023921
c_{14}	11769783413683776	c_{39}	−4781688671532655102588354895544320	c_{64}	−708912556118513
c_{15}	−87199437041525888	c_{40}	−4542027069261546233617010403573760	c_{65}	39916782103963585
c_{16}	−702967187710820352	c_{41}	48288111876059836871158791034372096	c_{66}	9016894704688520
c_{17}	5249640954289870080	c_{42}	38234909555102435061940139467997184	c_{67}	−8103418403747018
c_{18}	34575580199575725056	c_{43}	−430900326506991308345633495140270080	c_{68}	−672473984310288
c_{19}	−261061625320699761152	c_{44}	−282834364542156170101111312361717760	c_{69}	14379963190026332
c_{20}	−1419136466144658165248	c_{45}	3402151862513382467010785310897340416	c_{70}	−341714774753236
c_{21}	10865988127678351005696	c_{46}	1838466840426563169791652539108687872	c_{71}	−2221606003553075
c_{22}	49123895665615873411072	c_{47}	−23789406892060430849588711409133289472	c_{72}	2219791981023001
c_{23}	−382556950992852155645952	c_{48}	−10494752267591569189964092601492570112	c_{73}	29738001392572261
c_{24}	−1446441274117644068081664	c_{49}	147412890735019989295528425707079204864	c_{74}	−4477250955051242
c_{25}	11491312349072420866129920	c_{50}	52545794476482097227563119261933109248	c_{75}	−3429441339248534

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A233679 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 1$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{99} - c_1 x^{99-1} - \cdots - c_{99}$. Its constant term is $-c_{99} = 5316911983139663491615228241121378304$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 99 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 99, so $\deg q = 99$, and it is monic. It holds from some index t on. From $\max(t, 1)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 99, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 99, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is $-53169119831396634916152282411213783 04$.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-99 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A233679.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.